

- Introductory Material
- Introduction to Public Key Cryptography
- GCD, Euclid's Algorithm and Bezout Coefficients
- Lab: Notes on GCD, Euclid's Algorithm and Extended GCD

1h
- Video: Euclid's Algorithm and GCD

12 min
- Graded Assignment: GCD and Euclid's Algorithm

Submitted
- Video: Extended Euclid Bezout Coefficients

18 min
- Graded Assignment: Bezout Coefficients

20 min

RSA Public Key Cryptosystem

Basic Introduction to Quantum Computation

Problem Set: Week 1

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Bezout Coefficients

Next item →

Review Learning Objectives

1. 11 and 19 are relatively prime. Which of the following are valid Bezout coefficients (s, t) such that

2 / 2 points

$11s + 19t = 1$

☒ (7, -4)

☐ (3, 2)

☐ (-7, 4)

☒ Correct
 $7 \times 11 - 4 \times 19 = 77 - 76 = 1$

Assignment details

Due
May 5, 11:59 PM IST

Attempts
Unlimited

Submitted
Apr 30, 12:45 PM IST

Try again

Your grade

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

2. Suppose Alice and Bob have unlimited coins of 24¢ and 32¢. Alice wishes to pay 4 cents to Bob. Is there a way for them to exchange coins so that the bottomline result is that Alice pays 4 cents to Bob.

1 / 1 point

- ☐ Yes, Alice can pay five 24 cent coins to Bob and Bob pays back 4 32 cent coins.
- ☒ No, any number of the form $24s + 32t$ where s, t are integers must be divisible by 8.
- ☐ Yes, Alice can give Bob a three 32 cent coins and Bob gives her four 24 cent coins back.

☒ Correct
Correct

3. What is the smallest integer k such that

1 / 1 point

$15k \pmod{21} = 1$?

☒ No such integer since $15k \pmod{21}$ must be divisible by 3 for all k .

☐ $k = 7$

☐ No such integer since $15k \pmod{21}$ must be divisible by 9 for all k .

☒ Correct
Correct by Bezout theorem, $15k - 21q$ is divisible by 3 for all integers k, q .

4. If (s, t) are Bezout coefficients associated with m during execution of Euclid's algorithm and

3 / 3 points

(s', t') are associated with n , then what are the coefficients associated with $m \pmod{n}$? Assume

$m = qn + r$ where $r = m \pmod{n}$ and q is the quotient of m/n .

☒ $s \leftarrow s - qs', t \leftarrow t - qt'$

☐ s, t remains unchanged.

☐ $s \leftarrow t, t \leftarrow s$

☐ $s \leftarrow s - qt, t \leftarrow t - qs$

☒ Correct
Correct: because $r = m - qn$